Curriculum Vitae

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EDUCATION

- **Tsinghua University** 08/2017 – 07/2021(Expected)
  - Bachelor of Engineering, Department of Automation
  - Accumulative **GPA**: 3.73/4.0 (3.86/4.0 for the junior year)
  - **Awards**: Friends of Tsinghua - Guangzhou Pharmaceuticals Scholarship, Academic Excellence Scholarship

PUBLICATION

- Zhuolin Yang*, **Zhaoxi Chen**, Tiffany Cai, Xinyun Chen, Bo Li, Yuandong Tian*. *Understanding Robustness in Teacher-Student Setting: A New Perspective.* Accepted at AISTATS 2021.
- **Zhaoxi Chen***, Xiang Zhang*, Yuwang Wang, Zhen Fan, Yipeng Li, Xiu Li, Qionghai Dai. *FlyingCap: Multi-Agent Deep Reinforcement Learning for Active 3D Human Pose Estimation.* Under review for CVPR 2021.
- Zhong-Han Niu, **Zhaoxi Chen**, Linyi Li, Yubin Yang, Bo Li, Jinfeng Yi. *On the Limitations of Denoising Strategies as Adversarial Defenses.* Under review for CVPR 2021.

ACADEMIC EXPERIENCE

- **Department of Automation, Tsinghua University** 12/2020 – now
  Diploma thesis supervised by Prof. Gao Huang

  Dynamic Approaches to Reduce Redundancy in Video Classification

  - On-going research.
  - Investigated spatial and temporal redundancy of time-sequential data.
  - Proposed dynamic approaches to reduce spatial and temporal redundancy in video classification.

- **Secure Learning Lab (SL2), Computer Science Department, UIUC** 06/2020 – 11/2020
  Intern advised by Prof. Bo Li

  Understanding Robustness in Teacher-Student Setting: A New Perspective

  - Leveraged the teacher-student framework to understand adversarial examples of models.
  - Proposed principles about the relationship between normalized correlation (indicator of student specialization), robustness and the cause of adversarial samples.
  - Conducted experiments extensively to verify our analysis in different scenarios, including standard training, adversarial training, training with robust feature, etc.
  - Paper accepted at AISTATS 2021 as the second author.

  On the Limitations of Denoising Strategies as Adversarial Defenses

  - Proposed a symmetric transformation framework (*F-IF Framework*) to understand existing denoising-based adversarial defenses.
  - Conducted experiments extensively to reveal the limitations of denoising strategies.

- Completed a research paper as the second author and submitted to CVPR 2021.

- **ModLab, GRASP, University of Pennsylvania** 05/2020 – 09/2020
  Intern advised by Prof. Mark Yim

  Active Reconfiguration with SMORES-EP, a Modular Robot

  - Proposed a graph neural network(GNN) based framework for configuration identification of modular robots consist of a variable number of modules.
  - Addressed the problem of lack of training data from a perspective of few-shot learning.
  - Developed an *Unity* based simulator for modular robot, SMORES-EP, where user can construct, verify and simulate their configuration designs of modular robots.

- **Microsoft Research Asia** 03/2020 – 10/2020
  Research Assistant advised by Dr. Yuwang Wang

  Multi-Agent Deep Reinforcement Learning for Active 3D Human Pose Estimation

  - Became the first to address active 3D pose estimation problem with a distributed multi-agent based deep reinforcement learning framework.
  - Applied kinematic constraint and inter-agent communication to maintain realistic camera motions.
  - Added value prediction module to policy network to predict future movements of cameras.
  - Illustrated the feasibility of the system in a simulation environment.
  - Generated a research paper as the first author and submitted to CVPR 2021.

- **BBNC Lab, Department of Automation, Tsinghua** 08/2018 – 09/2020
  Research Assistant advised by Prof. Yipeng Li

  "Wild Array" Collaborative UAV Intelligent System

  - Led a team of nine to develop a collaborative intelligent UAV system consists of 3 UAVs.
  - Constructed the system from scratch, including body design, system hardware architecture and multi-sensor fusion.
  - Tested algorithms including autonomous positioning, 3D mapping, human body detection and swarm path planning with the system.
  - Optimized the visual-inertial odometry VINS-Mono on TX2 for real-time use.
  - Applied the system into Mohamed Bin Zayed International Robotics Challenge 2020.

ACTIVITIES

- **The 17th Tsinghua Siyuan Excellent Student Training Program** 05/2018 – Present
  - Selected 35 out of 3300 students.
  - Training in leadership skills and social responsibility.
  - Provided volunteer education for Shangluo, Shaanxi, China.

- **Saxophone Performer – Tsinghua University Student Art Troupe** 09/2017 – Present

ENGLISH PROFICIENCY AND OTHER SKILLS

- **English Proficiency:** TOEFL 107 (MyBest); GRE 158+170+4
- **Programming Skills:** Proficient in C/C++, Python, Java, C#, MATLAB, Verilog, LaTeX
- **Software & System:** Linux, ROS, Git, Unity, OpenCV, PyTorch, Numpy, Altium Designer